

data_curve_fit

June 23, 2025

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[1]: import matplotlib.pyplot as plt
import numpy as np
from astropy.io import fits
from read_L1642_data import *
from scipy.optimize import minimize
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[2]: #read L1642 data
#dat = read_L1642_data(band='u')
def chisqrt(x, arg):
    band = arg
    dat = read_L1642_data(band=band)

    # Wavelengths
    # u,      b,      y,      385,      415
    # 3500.0, 4670.0, 5550.0, 3840.0, 4160.0
    Aobs = dat.Aobs
    Aobs_err = dat.Aobs_err
    Iobs = dat.Iobs
    Iobs_err = np.amax([dat.Iobs_err1, dat.Iobs_err2], axis=0)
    #Iobs_err = (dat.Iobs_err1 + dat.Iobs_err2)/2.0
    #Iobs_err = np.sqrt(dat.Iobs_err1**2 + dat.Iobs_err2**2)/2.0

    #w = np.where(Aobs > 0.5)[0]
    #Aobs = Aobs[w]
    #Aobs_err = Aobs_err[w]
    #Iobs = Iobs[w]
    #Iobs_err = Iobs_err[w]
    #Aobs = Aobs[:-2]
    #Aobs_err = Aobs_err[:-2]
    #Iobs = Iobs[:-2]
    #Iobs_err = Iobs_err[:-2]

    I0 = x[0]
    C1 = x[1]
    C2 = x[2]
    tau = Aobs / 1.086
    Imod = I0 * np.exp(-C1*tau)*(1.0-np.exp(-C2*tau))
    chi2 = np.sum((Iobs - Imod)**2/Iobs_err**2) / (Iobs.size - 3.0)
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    return chi2

def plot_model_out(band):
    method = 'Nelder-Mead'
    #method = 'Powell'
    x0      = [61.0, 0.1, 0.5]
    bounds = ((0.0, None), (0.0, 1.0), (0.0, 1.0))
    result = minimize(chisqrt, x0, args = (band), bounds=bounds, method=method)
    #result = minimize(chisqrt, x0, args = (band), method=method)
    x       = result.x
    chi2    = result.fun

    I0      = x[0]
    C1      = x[1]
    C2      = x[2]

    xmax    = 1.0/C2*np.log(1.0+C2/C1)
    ymax    = I0 * (C1/(C1+C2))*(C1/C2)*C2/(C1+C2)
    print(x, chi2, result.success)
    print(xmax, ymax)

    tau     = np.arange(0.0, 15.0, 0.1)
    Aext    = tau * 1.086
    Imod    = I0 * np.exp(-C1*tau)*(1.0-np.exp(-C2*tau))
    dat     = read_L1642_data(band=band)

    lab     = r'$I=%4.1f\, e^{-%4.3f\tau}(1-e^{-%4.2f\tau})$' % (I0, C1, C2)
    plt.plot(Aext, Imod, label=lab)
    plt.errorbar(dat.Aobs, dat.Iobs, xerr=[dat.Aobs_err], yerr=[dat.Iobs_err1,
↳ dat.Iobs_err2], fmt='o', markersize=4.0, color='black')

    plt.legend(frameon=False)
    plt.xlim(0.0, np.amax(dat.Aobs)*1.1)
    plt.ylim(0.0, np.amax(dat.Iobs)*1.1)
    plt.xlabel(r'$A_{\rm %s}$' % band)
    plt.ylabel(r'$I_{\rm %s}$' % band)

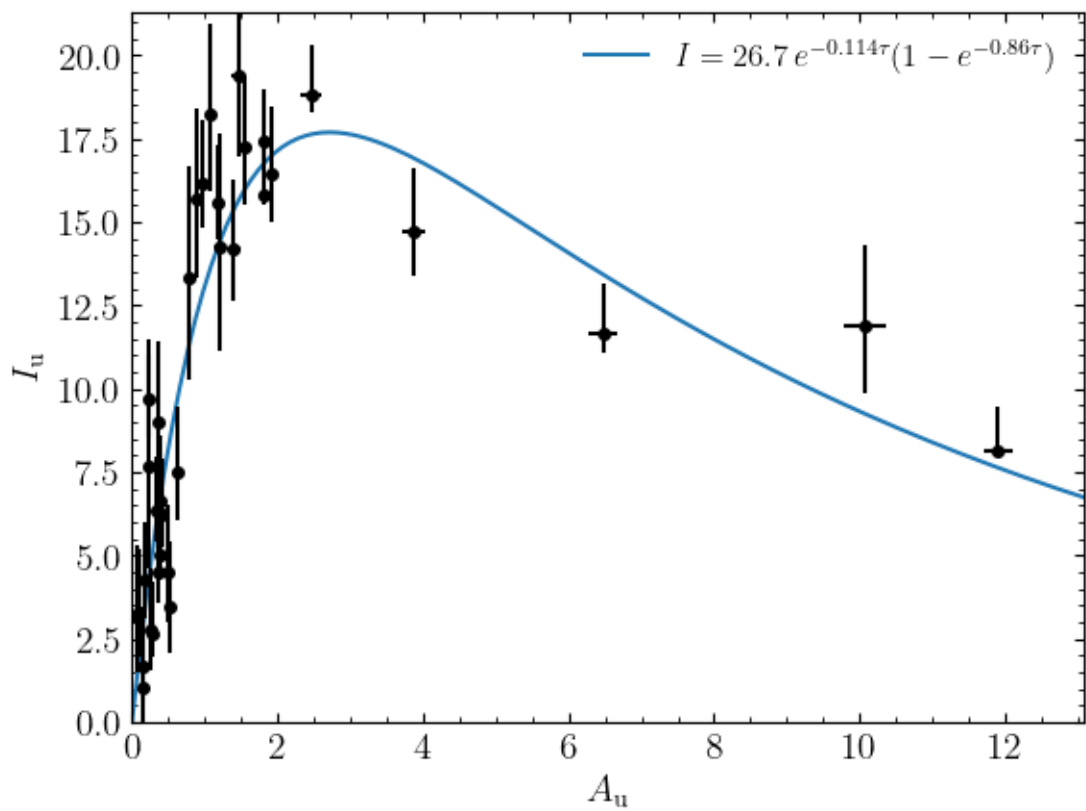
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[3]: plot_model_out('u')
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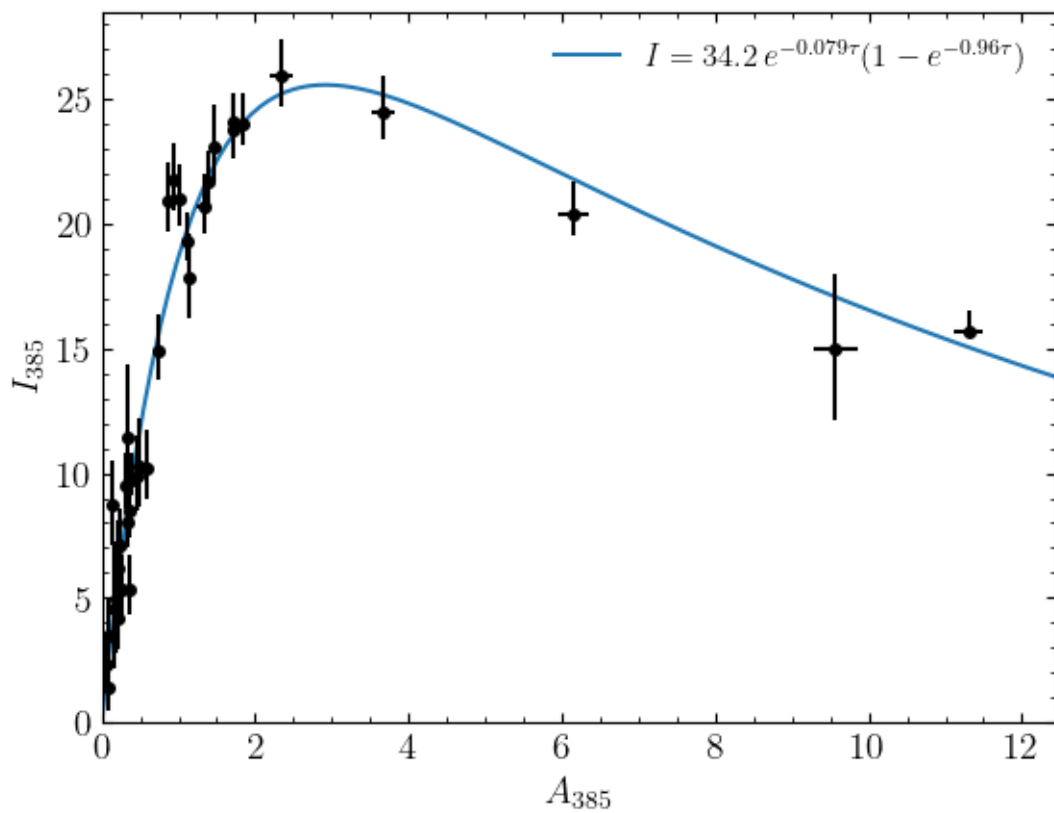
[26.67674647  0.11411131  0.85543998] 1.5033349954975674 True
2.5012379687849298 17.692755553896887

```



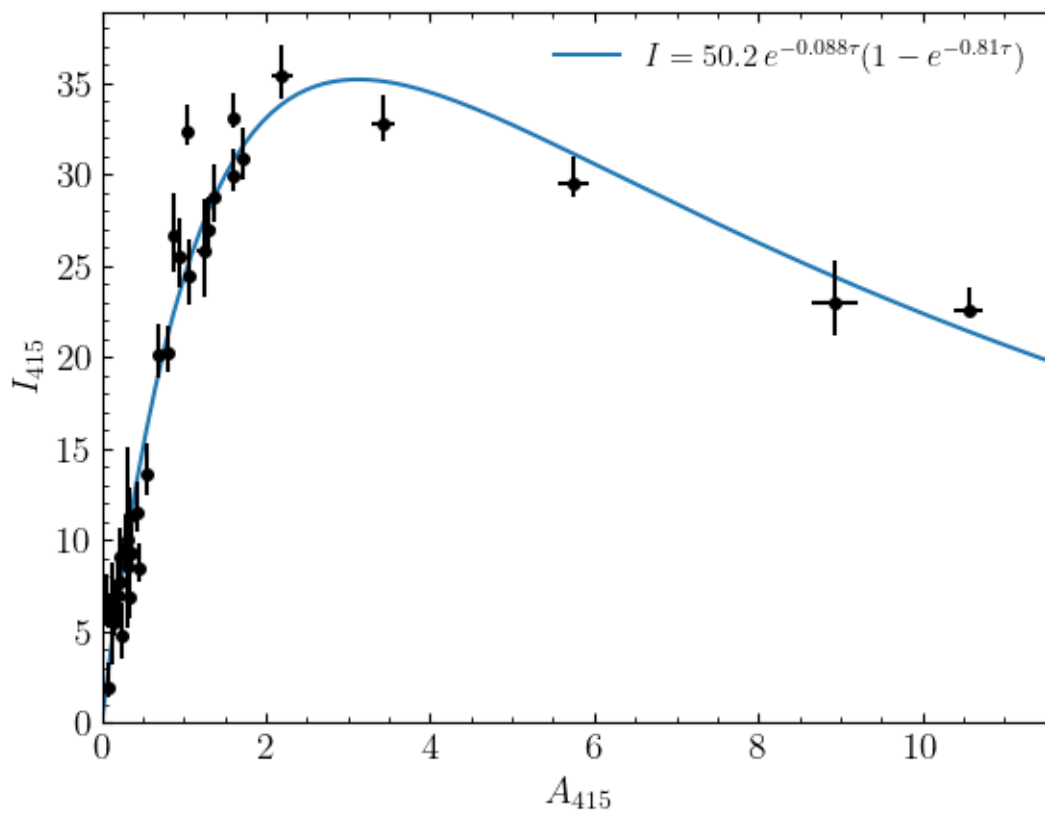
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[4]: plot_model_out('385')
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[34.19422753  0.07884309  0.96110379] 1.4573423303355482 True
2.6838571358342653 25.57488341142097
```



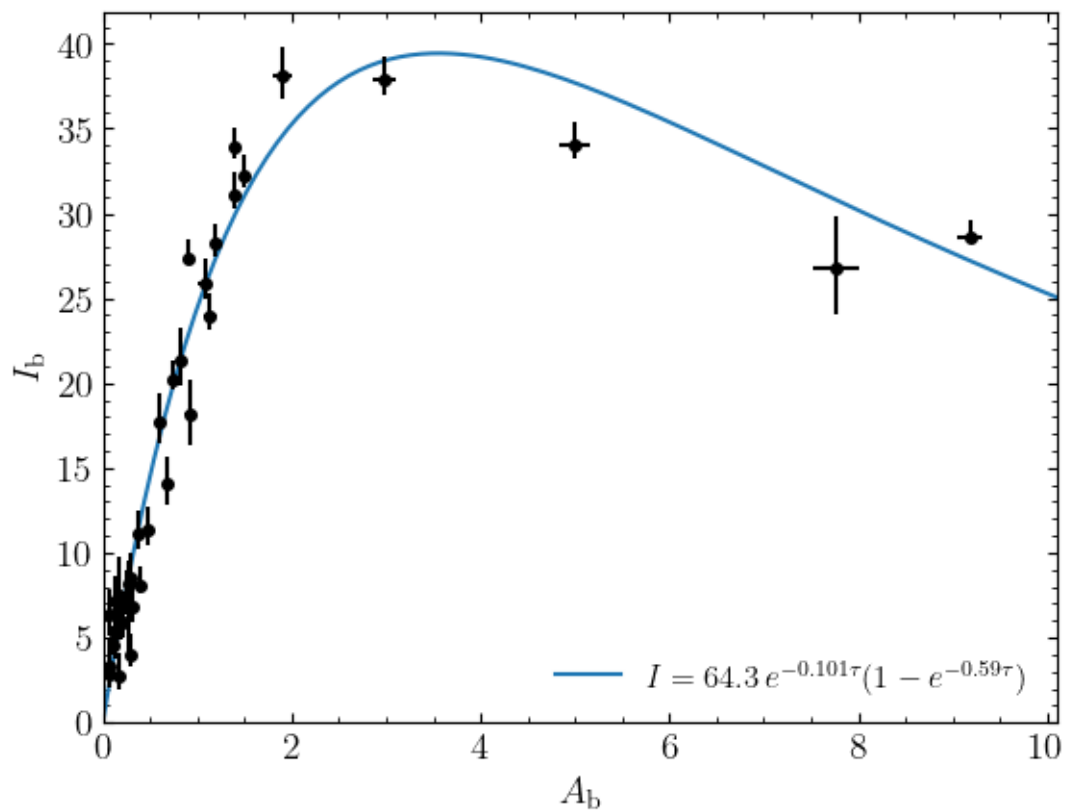
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[5]: plot_model_out('415')
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```
[50.15693297  0.08752803  0.80925851] 2.633382820610393 True
2.8752972892920856 35.19092345232604
```



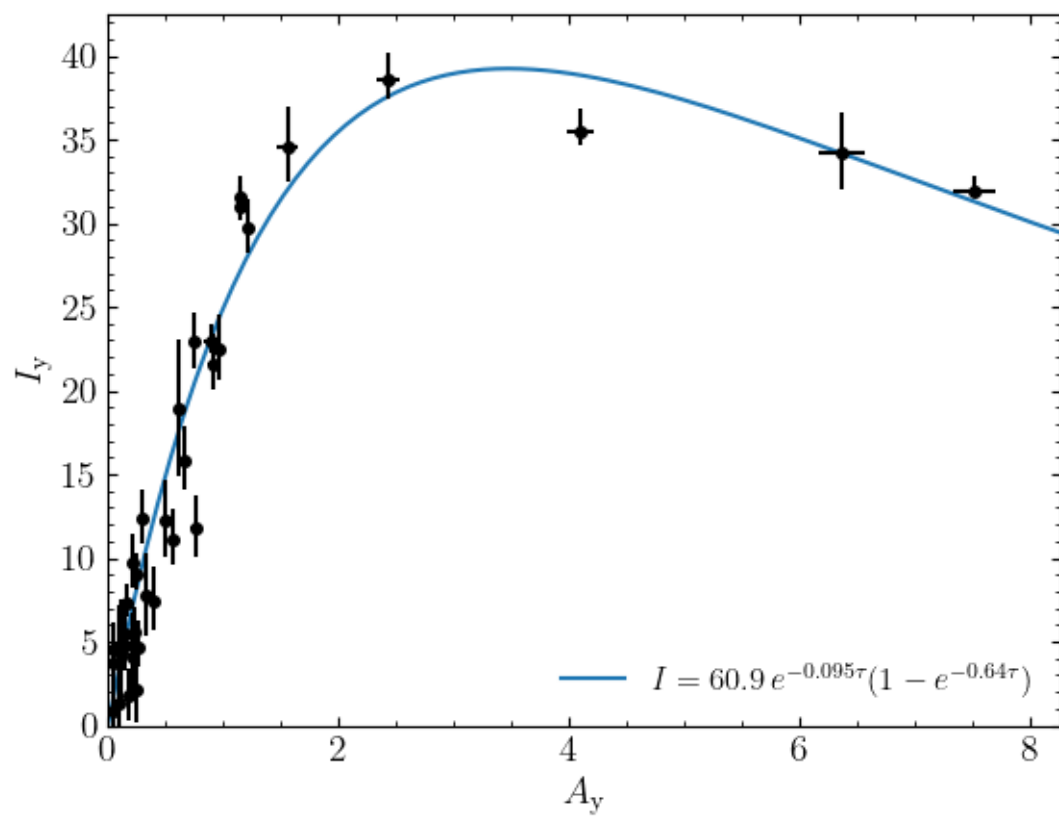
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[6]: plot_model_out('b')
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[64.28711851  0.10083154  0.58656918] 3.707879019413793 True
3.272361110684752 39.43979182602296
```



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[7]: plot_model_out('y')
```

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[60.9325458  0.09460337  0.64124342] 3.4027377809684882 True
3.1989862219539935 39.23306122465626
```



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